Open Source Software Project Report

Project Title: Open Source Audit

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Course: Open Source Software

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Introduction

Open Source Software plays a very important role in the field of computing. Many applications we use every day are based on Open Source technology. Open Source Software is the software which provides the freedom to view, modify, and share the source code.

In this project, I have selected Git as the Open Source Software. Git is the Open Source Software which is used by the programmers to maintain the versions of the source code.

The aim of this project is to understand Open Source philosophy, Open Source license, Linux environment, and shell scripting. I have also developed five shell scripts to show the usage of Linux command.

PART A — Origin and Philosophy

A1 — Problem Git was created to solve

Git is the Open Source Software which is developed by Linus Torvalds in the year 2005. Before the invention of Git, many programmers were using other version control tools which were very slow. Linus Torvalds needed to develop a fast and efficient version control tool to maintain the Linux kernel. Many programmers were working together to develop the Linux kernel. It is very difficult to maintain the versions of the source code because many programmers were working together.

Git is developed to overcome the above difficulties. Git provides the freedom to many programmers to work together on the same project. Git provides the flexibility to maintain the versions of the source code. Git is the Open Source tool which is freely available. Many programmers contribute to the development of Git.

A2 — License of Git

The license for Git is the GNU General Public License. This license allows users to use, modify, and distribute the software for free. The four freedoms of open-source software are:

Freedom to run the program for any purpose

Freedom to study how the program works

Freedom to modify the program

Freedom to distribute copies

The GPL license ensures that the modified version of the software is also open-source.

Differences between GPL and MIT License:

The GPL license ensures that the modified version of the software is open-source, whereas the MIT license allows for private modifications.

What is Free Software?:

The free software means that users have the freedom to use and modify the software, not necessarily that it is free.

A3 – Ethics of Open Source:

The open-source software allows for the sharing of knowledge. Most companies use open-source software for developing applications.

Advantages:

Promotes innovation

Saves money

Improves security

Disadvantages:

Companies use open-source software without giving anything in return

No support from developers

The open-source software allows developers to learn from the open-source code. This helps in improving the quality of the software.

PART B – Linux Footprint of Git:

The Git software is usually installed in Linux. The command for installing the Git software is:

sudo apt install git

The Git software is usually found in the following locations:

/usr/bin/git

/etc

/home

The Git software does not require any special permissions. It can run on normal user permissions. The updates for the Git software are usually provided by the open-source community. The users of the Linux operating system can easily update the Git software by typing commands in the terminal.

PART C – FOSS Ecosystem:

The Git software is usually used in combination with other open-source software such as:

GitHub

GitLab

Linux

VS Code

Git facilitates collaborative software development. It is commonly used as a version control system in open source projects.

It is a part of modern software development ecosystem and helps in team work.

PART D — Open Source vs Proprietary Software

Feature	Git (Open Source)	Proprietary Software
Cost	Free	Paid
Security	Code visible	Code hidden
Flexibility	High	Limited
Community	Large	Limited
Customization	Allowed	Restricted

Conclusion:

It is highly recommended as it is free, reliable, and supported.

SCRIPT SECTION

Paste code + screenshot for each script.

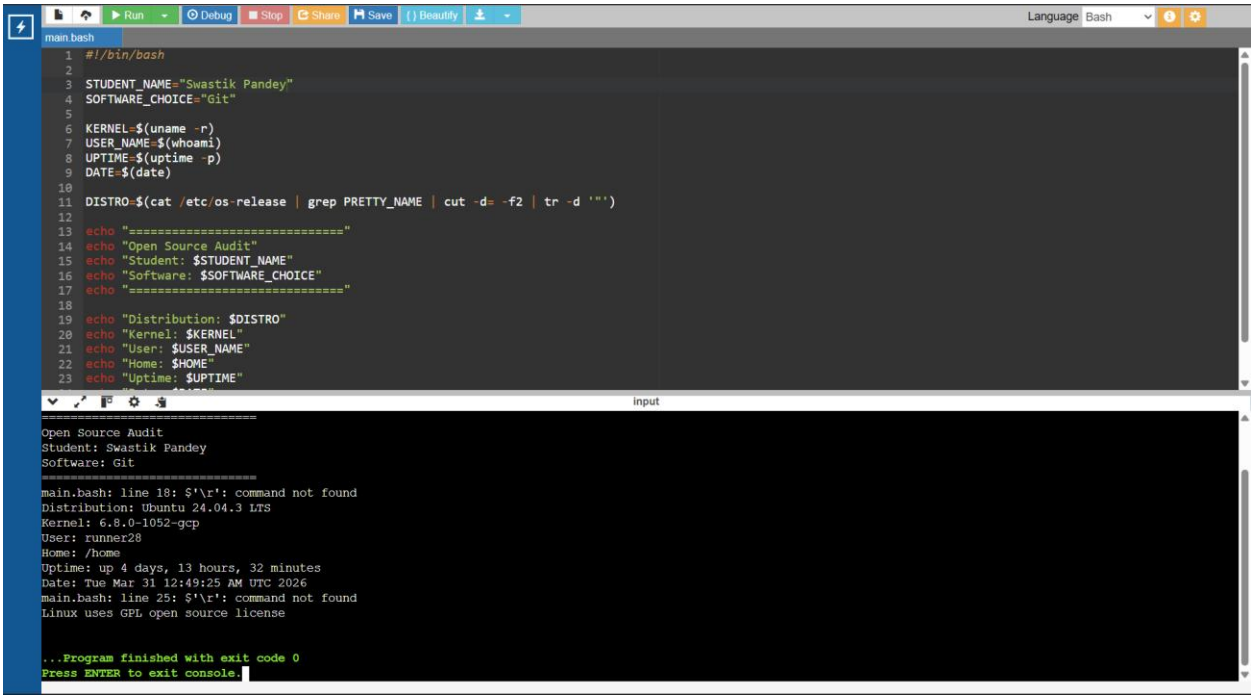
Script 1 — System Identity Report

Code

```
#!/bin/bash

# Script 1: System Identity Report
# Author: YOUR NAME

STUDENT_NAME="YOUR NAME"
SOFTWARE_CHOICE="Git"

KERNEL=$(uname -r)
USER_NAME=$(whoami)
UPTIME=$(uptime -p)
DATE=$(date)
```

```
DISTRO=$(cat /etc/os-release | grep PRETTY_NAME | cut -d= -f2 | tr -d '"')

echo "========================================"
echo " Open Source Audit — $STUDENT_NAME"
echo "========================================"

echo "Software Chosen : $SOFTWARE_CHOICE"
echo "Distribution    : $DISTRO"
echo "Kernel Version  : $KERNEL"
echo "User            : $USER_NAME"
echo "Home Directory  : $HOME"
echo "Uptime          : $UPTIME"
echo "Current Date    : $DATE"

echo ""
echo "License Info: Linux kernel is released under GPL license"
```

Screenshot



Explanation

This script will show Linux system information such as username, kernel version, system uptime, and system date using basic shell commands.

Script 2 — FOSS Package Inspector

Code

#!/bin/bash

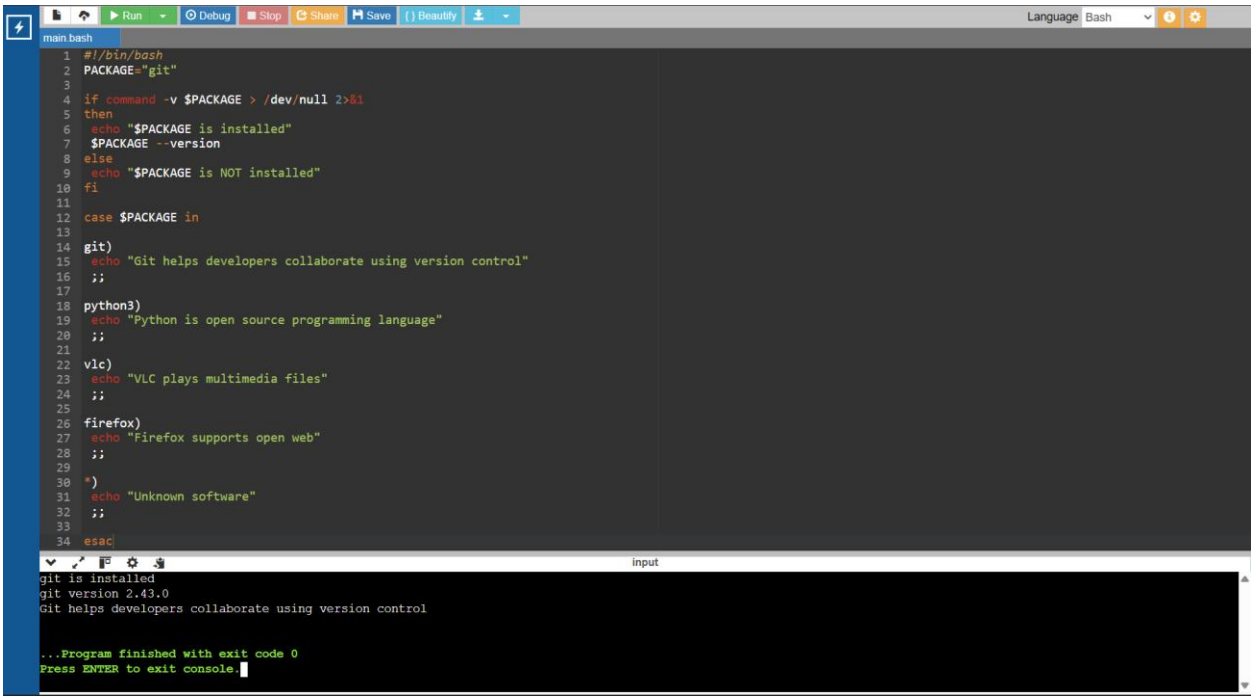
Script 2: FOSS Package Inspector

```
PACKAGE=${1:-git}

echo "Checking package: $PACKAGE"

if dpkg -l | grep -qw $PACKAGE; then
   echo "$PACKAGE is installed."

   dpkg -s $PACKAGE | grep -E 'Version|Maintainer'

else
   echo "$PACKAGE is NOT installed."
fi

case $PACKAGE in

git)
echo "Git: Distributed version control system promoting collaboration"
```

```
;;

python3)
echo "Python: community-driven programming language"
;;

vlc)
echo "VLC: open media player built by volunteers"
;;

firefox)
echo "Firefox: browser supporting open web standards"
;;

*)
echo "Unknown package"
;;
```

esac

Screenshot



Explanation

This script will check if the software named 'git' is installed in the system or not. It will show version information and description of the software.

Script 3 — Disk and Permission Auditor

Code

```
#!/bin/bash

# Script 3: Disk and Permission Auditor

DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "--------------------------------"

for DIR in "${DIRS[@]}"

do

if [ -d "$DIR" ]

then
```

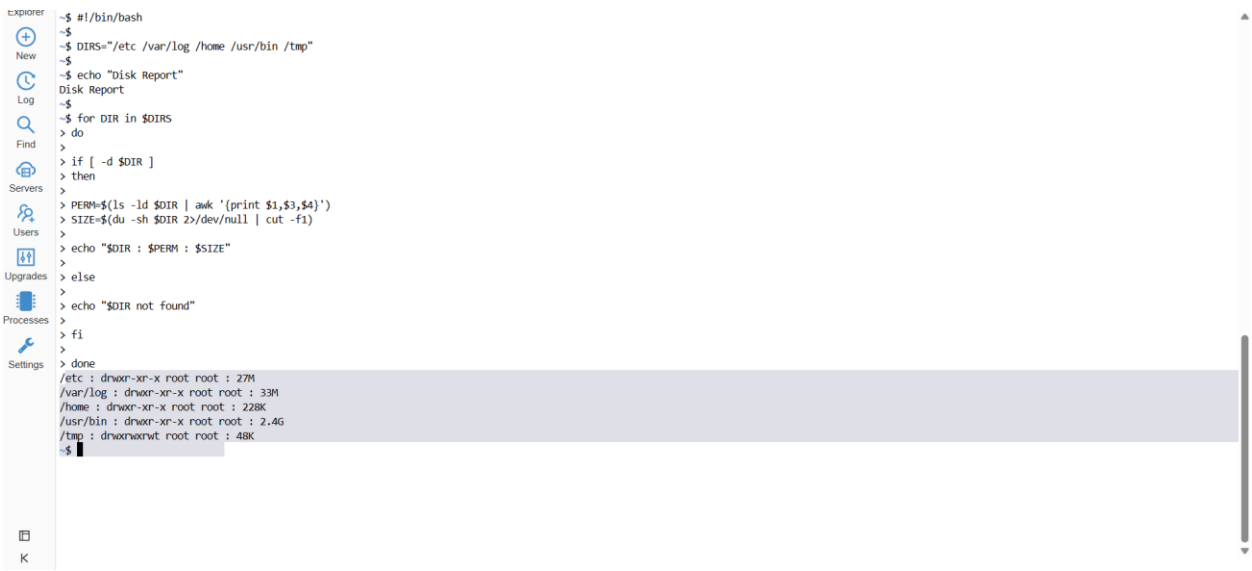
```
PERMS=$(ls -ld $DIR | awk '{print $1, $3, $4}')

SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)

echo "$DIR => Permissions: $PERMS | Size: $SIZE"

else

echo "$DIR does not exist"

fi

done


# check git config directory

CONFIG_DIR="$HOME/.gitconfig"
```

```
echo ""
echo "Checking Git config file"

if [ -f "$CONFIG_DIR" ]

then

ls -l $CONFIG_DIR

else

echo "Git config not found"

fi
```

Screenshot

```bash
#!/bin/bash

DIRS="/etc /var/log /home /usr/bin /tmp"

echo "Disk Report"

for DIR in $DIRS
do

if [ -d $DIR ]
then

PERM=$(ls -ld $DIR | awk '{print $1,$3,$4}')
SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)

echo "$DIR : $PERM : $SIZE"

else

echo "$DIR not found"

fi

done
```

```
Disk Report
/etc : drwxr-xr-x root root : 27M
/var/log : drwxr-xr-x root root : 33M
```



Explanation

This script will show directory permissions and disk usage information using loop and Linux commands to analyze system directories.

Script 4 — Log File Analyzer

Code

```
#!/bin/bash

LOGFILE="/var/log/syslog"
KEYWORD="error"

COUNT=$(grep -i $KEYWORD $LOGFILE | wc -l)

echo "Total errors: $COUNT"

echo "Last 5 lines:"
grep -i $KEYWORD $LOGFILE | tail -5
```

Screenshot

Explanation

This script will read a log file and show the number of errors in the file. It will help in identifying system issues using grep and counting commands.

Script 5 — Open Source Manifesto Generator

Code

```
#!/bin/bash

# Script 5: Open Source Manifesto Generator

echo "Answer the following questions"
```

```
read -p "1. Name open source tool you use daily: " TOOL

read -p "2. One word meaning of freedom: " FREEDOM

read -p "3. Something you want to build and share: " BUILD


DATE=$(date)

OUTPUT="manifesto_$(whoami).txt"


echo "On $DATE, I believe open source represents $FREEDOM." > $OUTPUT

echo "Tools like $TOOL show how collaboration helps everyone." >> $OUTPUT
```

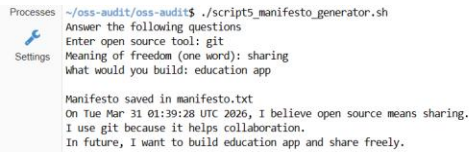
echo "In future, I want to build $BUILD and share it freely with the community." >> $OUTPUT

echo ""

echo "Manifesto created: $OUTPUT"

cat $OUTPUT

Screenshot



Explanation

This script will generate a personal open source manifesto using input from users and store it in a text file.

Conclusion:

In this project, I learned open source software, Linux environment, and shell scripts. Git is a powerful open source tool used worldwide, and writing shell scripts helps in understanding Linux commands.

Open source software plays a major role in software development.

Final Step

Save document as:

OSS_Project_Report.pdf

GitHub repo link:
https://github.com/swastik24bai10957-beep/oss-audit-24BAI10857